

Connect to the FPGA testbench host computer

We connect this Nexys4 FPGA development board to a host computer in the lab.



To log onto the computer, connect to eduroam, or princeton VPN if off-campus, then:

```
ssh -L 5905:localhost:5905 bench@10.50.62.45
```

The password is

FPGAtest1

Do not distribute.

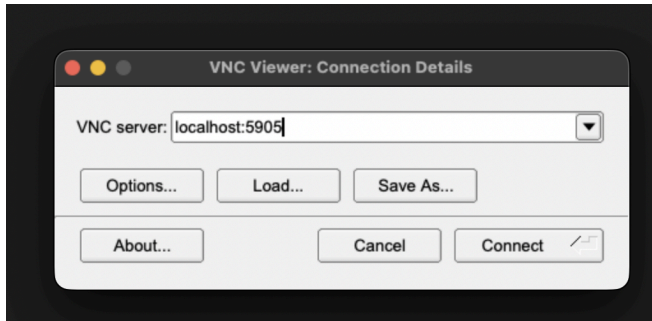
The file system on the host computer contains the following directories:

~/riscv-fpga	vivado source and project.
~/ece475-lab4	course project source.

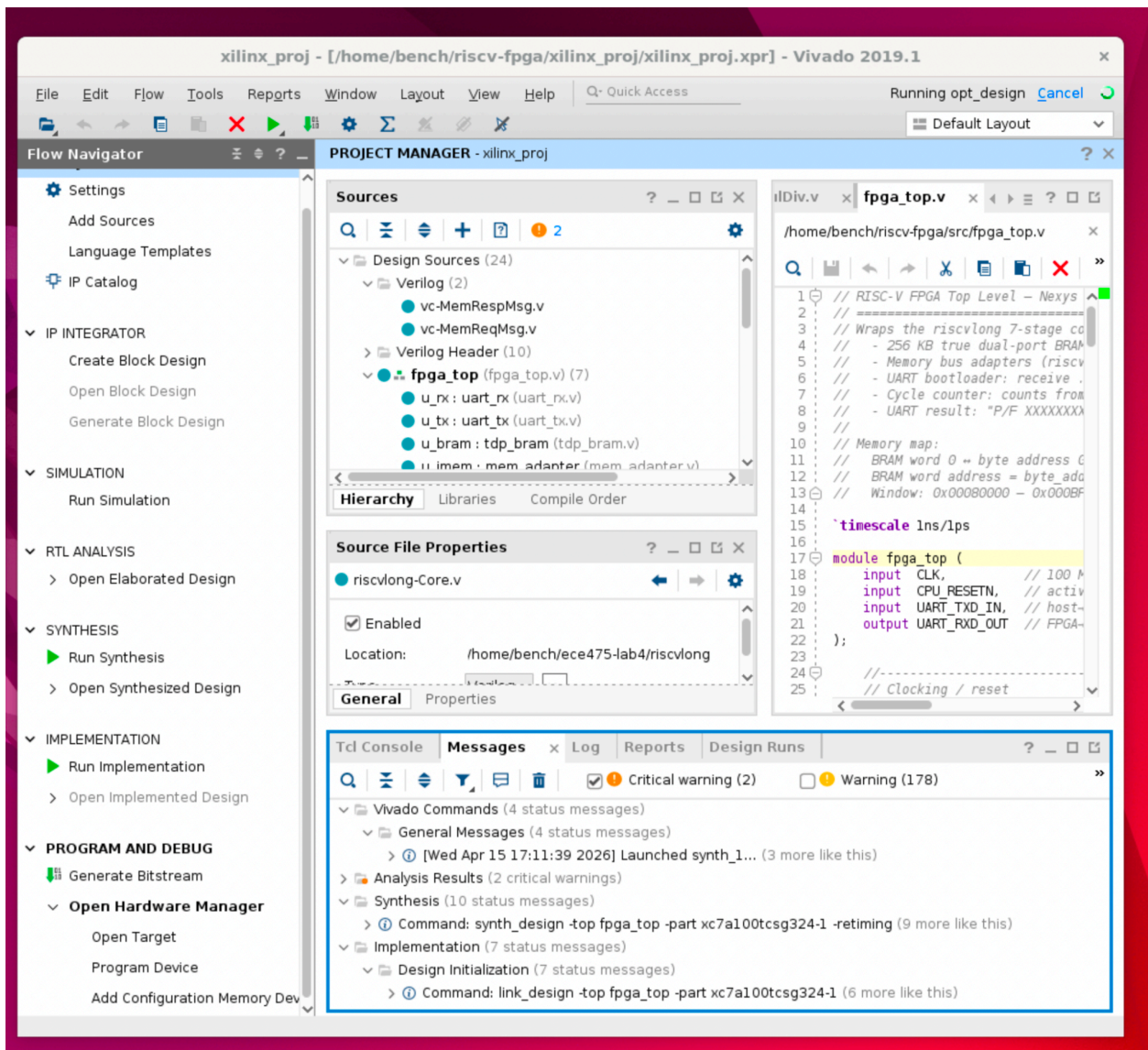
The riscvlong processor under ece475-lab4 is referenced by the vivado project and programmed onto the FPGA. You wish to edit the files under ece475-lab4 to experiment with your processor design.

Use Vivado to program FPGA

To use Vivado on the host computer through its graphical interface, you'll need to install a vncviewer like tigervnc <https://tigervnc.org/> on your computer. Once installed and while the ssh connection is live, connect to the forwarded port under localhost:



You'll enter the same password then see a desktop with Vivado already opened to the riscv-fpga project.



The standard flow here is (in the flow navigator): run synthesis, run implementation, generate bitstream, then program device.

Once programmed, you can use the terminal on the left to load a program vmh and obtain cycles back.

For some reason, probably the reset logic in the processor is not all correct, you'll have to reclick on program device before loading another program.

This computer does not have a riscv toolchain. To compile more programs you'll need to do it on adroit and copy them here.